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Coordination versus Prisoners' Dilemma: Implications for International Cooperation and Regimes

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The study of political institutions in general and international cooperation in particular has been beneficially influenced by the Prisoners' Dilemma (PD) game model, but there is a mistaken tendency to treat PD as representing the singular problem of collective action and cooperation. By relaxing the assumptions of 2 × 2 games and developing an alternate model of the coordination game, I show how some cooperation problems have very different properties from those found in PD. The analytical results of the two games are compared across several important dimensions: number of strategies available, number of iterations of the game, numbers of players, and the distribution of power among them. The discussion is illustrated with specific problems of international cooperation, and the implications of alternative cooperation problems for the formation and performance of international regimes are explored. The basic solutions for PD and coordination have divergent ramifications for the institutionalization, stability, and adaptability of regimes and for the role of hegemony in the international system. However, the coordination model does not replace the PD model but complements and supplements it as a way to understand the diversity of political institutions. These results are widely applicable to areas of politics beyond international relations.

The problem of international cooperation is essentially one of collective action applied to the particular circumstances of the international system. The formation of international regimes raises the same general issues that surround the development of political institutions for resolving collective action problems in various spheres of human activity. The nature of these institutions may range from formal organizations suggested by notions of legal contract and central government through a wide variety of intermediate institutions to the very informal methods of collective organization suggested by norms and conventions (Krasner, 1982a, p. 186). This considerable range of institutional variation has important consequences for regime formation (or expansion) in different international issue areas. It poses the fundamental question of what sorts of regimes are appropriate for resolving particular problems of collective action, while simultaneously it leads to a consideration of the sources of variation among international regimes. The nature and different sources of this variation in regimes will be the focus of this article. Although the discussion has

corresponding implications for the study of domestic politics and for political theory more generally, the analysis will be presented in terms of the problematique of "international regimes." The lack of preexisting centralized institutions which characterizes international "anarchy" highlights the differences among collective action problems and among their solutions, whereas the substantive questions of international politics serve to concretize the discussion.

One explanation of institutional variation lies in the possibility that different regime forms may provide different solutions to the same problem; that is, given any particular problem of collective action, a solution might be achieved in any number of different ways and be reflected in correspondingly different institutional development. For example, the problem of maintaining free trade among states might be resolved by formal contract and enforcement through international treaties and organizations. Alternatively, the same trade problem might be resolved through the emergence of norms and conventions that provide informal guides for state conduct. In fact, the current international trade regime seems to fall somewhere between these extremes (Lipson, 1982). Of course, these different solutions are likely to entail other important differences. More formal regimes may have advantages in preventing states from taking noncooperative actions for short-run benefit, whereas less well institutionalized regimes may be better in allowing states flexibility to adapt international norms to domestic needs. Despite

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these variations, the different regime possibilities and combinations are clearly different solutions to the same fundamental problem of maintaining a liberal free trade regime (with its presumably generalized benefits). In this way, the diversity among different regimes could be understood even if the underlying structure of all collective action problems were similar.

A second explanation for regime variation is that regimes in different issue areas represent different solutions to fundamentally different problems. Different histories and contexts of particular international issues may facilitate certain types of solutions in some issues and other solutions in other issues. For example, the historical development of the norm of open access to ocean transportation routes provides a solid foundation for an informal norm-based regime, whereas in international trade, discrimination against foreign goods is a well-recognized right of states, and so the norm of nondiscrimination is inadequate as the sole basis of a regime. But an even more important source of this type of regime variation lies in the possibility that the underlying strategic structure of issues is fundamentally different; this will occur when the nature of interdependence between states, as reflected in the ways in which they affect and are affected by one another, differs across issue areas. If this is the case, then there is no longer any single problem of collective action. Instead, multiple different problems of collective action need to be recognized. These multiple problems may be soluble by similar types of regimes, or they may each require different institutional forms for their resolution.

Both explanations of variation in regime type are important, but the impact of strategic structure is generally less well understood than are other sources of regime variation. In large part this is because a particular model of collective action, the Prisoners' Dilemma (PD), has incorrectly come to be treated as the problem of collective action. Indeed, sometimes even a very special case of PD—the public good problem—is treated as if it were the sole problem.¹ A wide range of issues have been analyzed this way without careful scrutiny to check that their underlying strategic structure corresponds to this particular formalization. Although PD (and public goods) provide an important category of collective action

problems, inappropriate analysis and erroneous conclusions result when other issues are treated as if they are PD when in fact they are not.² Other collective action problems hold very different implications for the likely nature and performance of regime institutions.

An illustrative example of this problem is found in the application of one of the central theorems of the international regimes literature: the theory of hegemonic stability (Keohane, 1980, 1984; Kindleberger, 1978). Although the theory is ultimately concerned with the dynamics of the world system, its analytical underpinnings are essentially static and based on Olson's (1965) analysis of "privileged groups" in public goods problems.³ The theory claims that 1) international systems with dominant states are likely to have stable regimes provided by that dominant state; and 2) this will be to the advantage of the smaller states which, in Olson's terms, will exploit the large states by free riding on this provision by the dominant state. The surprising implication is that weaker states benefit from international inequality while hegemonic powers are better characterized as charitable benefactors than imperialistic or exploitative masters. (The dynamic corollary is that the rise or decline of stable international regimes is a response to the rise or decline of a dominant state in a particular international issue.) These are striking claims both in their positive derivation and their normative implications. It is hardly surprising that they have been picked up by a large number of scholars and applied to a diverse range of international issues. Unfortunately, many of these applications have been inappropriate.⁴

The problem with the theory of hegemonic stability does not rest with Olson, because his analysis is correct for the situation he analyzes (i.e., public good provision). Instead, the problem

²There are important exceptions where different categories of collective action problems have been recognized. See especially Stein (1982), who makes various insightful points about the difference between what he calls "dilemmas of common interest" and "dilemmas of common aversion." His analysis parallels many of the points raised below. For a more analytical treatment see Runge (1981) and Sen (1967).

³A group is privileged if at least one member receives sufficient benefit from the public good that he is willing to provide the good himself. Strictly speaking, the privileged group situation is not PD in the range of provision up to that provided by the largest group member. However, the level of provision is still suboptimal, and a PD dilemma persists with regards to reaching the collectively optimal level of provision (Snidal, 1981, chap. 3).

⁴See Snidal (1985) for a more detailed elaboration and analytical critique of hegemonic stability theory.

¹In large part, this may be due to Olson's (1965) persuasive labelling of his analysis of public goods problems as "*The Logic of Collective Action*" (emphasis added). For an analysis of public goods as a subset of PD see Hardin (1971). Below I consider other problems that are equally problems of collective action but involve neither PD nor public goods.

must be parcelled out between the fuzziness and the inappropriateness with which the theory is put forward. First (the fuzziness), it is not clear exactly what is being provided. Much of the discussion makes it seem as though provision of the regime itself (rather than cooperation in the issue area) is the primary concern. If this is the case, then the analysis may well be correct if regime institutions have the properties of a public good. But such an analysis seems to miss the point by concentrating on the relatively minor question of who will bear the costs of supporting the regime institutions rather than on the more important questions concerning the nature of those institutions (and the outcomes thereby determined). Moreover, it divorces the question of regime provision from a consideration of the characteristics of the underlying issue area. This point is directly related to the second point (the inappropriateness) that if the good being provided relates to the underlying issue area, then application of the theory requires that the issue area indeed involve a public goods problem. Many of the issues to which the theory has been applied do not seem to involve public goods. In such cases, it should not be expected to pertain.⁵

Therefore, there is good reason to doubt that the theory of hegemonic stability will apply to all international issue areas equally. I consider below a case where the theory does not apply and where different conclusions about the impact of hegemony emerge. This case is discussed as part of a more general investigation of situations of strategic interdependence that result in regime characteristics quite different from those derived under the assumption that the underlying situation corresponds to PD. It provides an understanding of the emergence of different regimes (in part) as a response to fundamentally different problems of international cooperation. As a result, many of the emerging conclusions from the international regimes literature (of which the theory of hegemonic stability is a prominent example) are less widely applicable than is generally believed.

The diverse structures of collective action problems will be analyzed in terms of two simple situations represented in game theoretic format—the

PD and coordination games—which pose fundamentally different problems for cooperation. Starting from the basic 2×2 representation of these games, the restrictive assumptions of the two-actor, two-choice, one-play game are each relaxed. The implications for (international) political institutions that emerge from these expanded games demonstrates that differences in the underlying strategic structures of international issues provides an important source of regime variation. The final section of the article brings these conclusions together in a discussion of regime institutionalization, regime stability, and the impact of changes in the distribution of power on regime outcomes. Significant differences between the two cases indicate the importance of recognizing that different problems of collective action may require different political resolution if international cooperation is to be successful.

Using Games to Study the Diversity of Collective Action Problems

Ordinal 2×2 games provide a useful way to investigate the variety and diversity of collective action problems. Although their use necessarily entails setting aside some contextual and historical richness, the models compensate by clearing away the underbrush of detail to facilitate clarity and rigor in our analysis. Of course they are particularly valuable in directing attention to the strategic aspects of interactions. But several of their major analytical assumptions are overly restrictive for the models to capture important aspects of real world situations. The assumptions of only two actors, each with only two choices and a single interaction between them need to be relaxed to consider a more realistic world of many states, each with multiple choices and repeated interactions.⁶ These alternative assumptions will be considered below in looking at the relationship of regime characteristics to the strategic structure of collective action problems.

⁶There are a host of other assumptions including, for example, that both players have perfect information concerning the game rules and payoffs, that ordinal payoffs provide an adequate description of the situation, that players are "rational" in the sense of being some sort of consistent maximizing agents, and that all relevant factors in the external game environment have been incorporated into the game. These assumptions may be heroic in certain circumstances and suggest the inadequacy of game models for understanding particular issues in isolation from substantive analysis of those issues. However, game models are useful for understanding that part of regime variation which can be accounted for in terms of differences in the underlying strategic structure across issues.

⁵No doubt artful reconstruction could make any of these issues seem to be public goods. For example, international cooperation itself can be argued to be a public good since (perhaps by definition) all parties are made better off. But this is only a sleight-of-hand which makes something seem to be a public good by distracting us from the actual properties of the underlying issue over which international cooperation is sought. Note that it could be used to show that all purely private goods market transactions are public goods.

Considering all possible combinations of ordinal preferences, there are 78 different 2×2 games which provide an exhaustive listing of the possible relations of interdependence in a two-state, two-choice, one-play world (Rapoport & Guyer, 1966). To investigate the implications of different types of strategic interdependence for regimes, I consider two of these: the PD and coordination games. Although the choice of these particular games may seem arbitrary (which in itself would not matter for showing that *some* different situations of interdependence lead to different implications for regimes), there are compelling reasons why a comparison of these two games is germane. First, these two games have been widely discussed as important problems both in the game theory literature and in substantive discussions of problems of international cooperation. Second (although it will not be developed here), in a typology of cooperation problems in 2×2 games, these two games emerge as the two polar cases of assistance and coordination problems as discussed below (Snidal, 1981, chap. 2). Third, they are analytically more tractable and fruitful than many other simple games.⁷

In this article I concentrate on regimes as ways to facilitate cooperation. As is evident below, this is one important force motivating the formation and subsequent development of regimes. However, there are also regimes where cooperation plays a limited role or even no role (e.g., where one state is able to impose a regime upon another) and the following discussion is not relevant to those cases. Also, the term "cooperation"—and the corresponding references to cooperative choices and outcomes—is used in the common-sense usage of working together for mutual advantage rather than in the misleading game theoretic sense of a cooperative game as one where binding agreements are possible. This latter piece of persuasion definition neatly assumes away most of the tough political problems involved in cooperation, especially with respect to the problematic of international anarchy, where the absence of a centralized capacity to enforce agreements is fundamental.

Finally, the use of game theoretic analysis implicitly entails a series of assumptions about the nature of international politics.⁸ Treating states as

goal-seeking actors with well-defined preferences implies an essentially realist (or neorealist) view of international politics. Aside from the usual dangers of anthropomorphism, such an analysis is incomplete insofar as either transnational or subnational actors have an independent role in international politics. Indeed, even the assumption that all states have choices to make that matter for one another is nontrivial. It may make the analysis more applicable to North-North or East-West issues than to many North-South issues where the preponderance of effect is unidirectional. Finally, I assume an international political environment where there is communication and sufficient order to make cooperation feasible. However, I do not assume strong central institutions able to enforce agreements strictly (although they are not in all instances completely powerless and, at a minimum, may serve to enhance the self-enforcement of international agreements). With all of these caveats—plus those that necessarily attend the use of simple models to understand a complicated world—I proceed.

The Assistance Problem: Prisoners' Dilemma

PD is well known as a problem of collective action and so is dealt with only briefly here. The normal (or strategic) game form is presented in Figure 1, where policy alternatives are indicated in bold face (0,1) for each of the two states, and the entries in each cell represent ordinal payoffs for States A and B, respectively. (The ordinal payoffs are represented as ranging from 4 = "Most Preferred Outcome" to 1 = "Least Preferred Outcome".)

PD is an archetypical example of the disjuncture between individual and group rationality which characterizes many problems of collective action: Pursuit of individual self-interest by states (represented by their each choosing their dominant strategy 1) results in their being worse off than if both abstain from pursuit of their narrow self-interest and cooperate (represented by their each choosing strategy 0). The dilemma persists even if cooperation is achieved, because both states will continue to have strong incentives to defect and the system is likely to return to the stable noncooperative and deficient equilibrium (i.e., outcome (2,2) in Figure 1).

The PD is the standard representation of externalities (including public goods) where in the pursuit of their own private gains actors impose costs

⁷The other compelling game which arises for these very same reasons is "Chicken." In terms of the typology of 2×2 games it is the case involving simultaneous assistance and coordination problems. It is discussed in Snidal (1981), but is excluded here because it is analytically more complicated than the other two models.

⁸The assumptions described here are not the only possibilities. For a discussion of game theory as a theo-

retical approach to international politics, see Snidal (in press).

Figure 1. Prisoners' Dilemma

		State B	
		0	1
State A	0	3, 3	1, 4
	1	4, 1	2, 2

on each other *independently of each other's action*; that is, in the pursuit of its national interest State A makes State B worse off regardless of what the latter does, and vice versa. Because each is affected more by the other's move than by its own move (e.g., State A's choice alters B's outcome by two ordinal rankings, whereas B's choice can affect B's outcome by only one ordinal ranking), they both need assistance from the other in the form of forbearance from its individually rational choice. Because each gains more by participating in a program of mutual assistance than by acting independently, there is room for cooperation in PD.⁹

The outcome of the single-shot PD depends heavily on the social-political context. If enforceable agreements are not possible, then neither state will cooperate, because each will be better off not cooperating regardless of what the other does. If binding agreements can be made, then both states will find it in their interests to enter into an agreement to cooperate. This notion of *contracting* out of PD depends on some equivalent to a Hobbesian sovereign able to enforce agreements. The sovereign might accomplish this by forcing states to perform on their promises or by imposing sanctions on failures to perform, so that the strategic structure is changed, and states find it in their self-interest to comply "voluntarily." Obviously such a sovereign is rarely present in international politics. However, as one extends the underlying analytic model beyond the single-shot PD, other factors appear that may allow for less centralized cooperation (e.g., iterated play) or otherwise affect the ease of solution by contract (e.g., bargaining problems in gradu-

⁹This is not true for all cases where one or both states need "assistance" from the other. For example, in the game of "Deadlock" (formed by interchanging the payoff pairs in the northwest and southeast cells of Figure 1), each is better off by foregoing the other's assistance rather than by accepting it at the cost of foregoing its own dominant strategy.

ated games). Investigating these possibilities requires relaxing the underlying assumptions.

Multiple Choices in the Graduated Prisoners' Dilemma

The assumption that states face only a dichotomous choice is a drastic simplification of the wider menu of choices typically available to states.¹⁰ A simple case in point is that of ocean fisheries, where the choices presented are not simply to fish or not fish any particular species but cover a complete range of intermediate choices expressible in terms of annual catches for each state (and species) involved. In other issues, the available policy choices may be more complex and multidimensional. For example, policy options regarding trade issues include not only tariffs and quotas but an array of techniques ranging from bureaucratic impediments to negotiation of voluntary export restraints and countertrading requirements. However, in relaxing the assumption of dichotomous choice, I assume that these many policy options (or combinations of them) can be arrayed meaningfully along a single dimension. Thus trade policy is treated as ranging from an open-door extreme to a closed-door extreme, with a full array of intermediate policy options in between.

In order to represent this full range of policy options, I use a graduated version of PD as shown in Figure 2. Note that the subgame defined by the two extreme strategies for each side and indicated by the four corner cells is identical to the simple 2×2 of Figure 1. Now five policy choices are shown for each state, while further intermediate policy choices could easily be inserted between any two policy options shown. The ordering of these policies (e.g., from "open" to "closed" in trade policy) is reflected in the ordering of the ordinal payoffs shown in the cells of the matrix. For any policy choice by State B, State A makes itself better off and State B worse off by choosing a less-cooperative (higher-numbered) strategy. Again, because the same is true of State B's choice given any choice by State A, there is room for

¹⁰Of course, there may be some cases where choices are strictly dichotomous or where states can deliberately structure the issue to reduce multiple choices to a single binary decision. The latter is especially relevant when threshold effects or focal points are involved. For example, "Mutual Assured Destruction" (MAD) as a weapons strategy is deliberately designed to avoid intermediate-sized nuclear wars by making nuclear war an all-or-nothing event. The examination of these binary choices has been a central theme in the work of Schelling (1960, 1978), which remains the authoritative source on these questions.

Figure 2. Graduated Form of Prisoners' Dilemma

		State A				
		.0	.25	.5	.75	1.0
State B	K	L				
	.0	3.0, 3.0	2.5, 3.3	2.0, 3.5	1.5, 3.8	1.0, 4.0
	.25	3.3, 2.5	2.8, 2.8	2.3, 3.0	1.8, 3.3	1.3, 3.5
	.5	3.5, 2.0	3.0, 2.3	2.5, 2.5	2.0, 2.8	1.5, 3.0
	.75	3.8, 1.5	3.3, 1.8	2.8, 2.0	2.3, 2.3	1.8, 2.5
	1.0	4.0, 1.0	3.5, 1.3	3.0, 1.5	2.5, 1.8	2.0, 2.0

Numerical entries should be treated as indicating only ordinal preferences where, for example, 3.5 is preferred to 3.25.

cooperation throughout the policy ranges available to the two states. This is a significant feature of graduated PD that every subgame (as defined by considering any two policy choices for each state) is also a PD. Thus the strategic structure of an issue is not altered by looking at gradations of strategy or, equivalently, *PD is the same in the small as in the large*.¹¹

Although the basic strategic structure of the game remains the same when intermediate policy options are introduced, the game is altered in two potentially important ways. First, a bargaining problem arises which is not present in the 2×2 version of PD. In the simple game there is a single cooperative and efficient outcome at the top left hand corner (3.3). But in the graduated game there is a larger set of available cooperative (and Pareto-efficient) outcomes as represented by the dark line segment $\overline{JKL}$ in Figure 2. Although agreement at (3.3) may seem compelling the way the game is set up, either state may attempt to bargain for a better outcome along $\overline{JKL}$, introducing new complications to impede cooperation.¹² A

second difference between cooperation in the simple and graduated games occurs when graduated policies allow states to achieve intermediate levels of cooperation in circumstances in which they are unwilling to risk complete cooperation. One or both states might be unwilling to risk full cooperation (strategy 1) because they cannot accept any possibility of receiving their worst outcome (i.e., a payoff of 1) in the event that the other state does not cooperate. In graduated PD, states may be able to achieve intermediate levels of cooperation (e.g., each play strategy .5) where the outcome if the other does not cooperate is not as disastrous (i.e., an outcome of 1.5).¹³ Such intermediate possibilities for cooperation also open up opportunities to learn cooperative behavior in a dynamic game setting. However, despite these differences, the important observation remains that the graduated PD preserves the basic structure of the simple PD game.

preferences are likely to obscure any "obvious" single solution. Technically, the difference rests with the fact that when the "core" contains only one point (as in Figure 1), it provides a compelling single solution, but when it has multiple cooperative points (as in Figure 2), it is indeterminate as to the precise outcome.

¹³This difference will not be significant in single-shot PD where any cooperation is unlikely—but it may be significant in iterated PD discussed below. In "Chicken" these graduated strategies are crucially important and change the nature of the game even in the single-shot game (Snidal, 1981, chap. 3).

¹¹Technically, this result follows because PD falls into the category of separable games where the impact of each player on the other can be assessed independently of what the other does. See Hamburger (1969) and Snidal (1981, chap. 3).

¹²The (3.3) outcome only looks compelling because the game is perfectly symmetric and information about payoffs and strategies is perfect. In real world situations, the lack of such a clear delineation of "most cooperative strategy" and differences in intensities of

Multiple States and Asymmetries in Prisoners' Dilemma

The extension of the two-actor PD to the n -actor situation is straightforward and well known; it corresponds to the standard case of (separable) externalities where each state imposes costs or benefits on others independent of their own behavior. In terms of ordinal preferences, every state is in the same symmetric relation to every other state (although this symmetry may break down under interval-level preferences where intensities are taken into account). This extension also poses no problem in the graduated game where it can be depicted in terms of an n -dimensional "cube" where each dimension represents one of the n state's policy alternatives and the outcome is represented by the point in the cube (and corresponding payoffs) determined by the choices of all n states. Standard examples include the "tragedy of the commons" where every state has an individual incentive to overexploit the commons even though this is contrary to the collective interests of all states.

The two standard results from n -actor PD are: 1) that larger numbers of states inhibit cooperation; and 2) that asymmetries (usually implicitly defined in terms of the impact of one state on another but, in fact, also a result of differences in preferences and intensities) facilitate cooperative and efficient outcomes. This latter result is often elaborated with the specific claim that smaller countries will benefit disproportionately in such asymmetric situations (i.e., the small exploit the large). However, these results are based specifically on public goods assumptions and are not valid as universal generalizations for all PD. Again, confusion of public goods as the problem of collective action is misleading for understanding the diversity of cooperation problems. But a precise specification of the conditions under which these generalizations apply does provide further insight into international cooperation. Not surprisingly, these conditions are closely related to the defining characteristics of public goods of "jointness" (i.e., that consumption by one state does not diminish the quantity or quality of the benefit available to the other) and, especially, of nonexclusion (i.e., that there is no way to prevent any state from receiving benefits).¹⁴ When

these conditions do not apply—and PD problems need involve neither jointness nor nonexclusion—then the generalizations need to be qualified or changed.

First, consider the claim that larger group size impedes cooperation. Insofar as this refers to the problems of organization that arise as a result of the different social-political features of large as opposed to small groups, the assertion is valid. As the number of members in any social grouping increases, there is likely to be a commensurate increase in problems of communication, in the tendency for states to perceive that their individual actions are less visible so that they can renege or cheat on cooperative arrangements without being noticed, and in the problems of resolving bargaining disputes over the distribution of benefits from cooperation. Even with an extensive and institutionalized apparatus for negotiation and diplomacy, states are not able to overcome such problems easily. Therefore, in general, multilateral negotiations involving greater numbers of states can be expected to increase the difficulty of cooperation.

However, an increased number of states also may have an opposite effect by opening up new areas (or levels) of potential cooperation which are not accessible to smaller numbers of states. There will be some projects on which no small group of states has incentive to cooperate but on which a larger group will find room for cooperation (e.g., projects such as the European airbus might not have been undertaken if, for example, there had been only two interested states). Even in public goods problems, an increased number of participants will not only increase the problems of organizing collective action, it will also increase the benefits from cooperation as reflected in the corresponding increase in the optimal level of provision. This beneficial impact of increased numbers of states may hold independently of any assumptions of jointness (e.g., where there are economies of scale in cooperation), but jointness will further enhance incentives for cooperation.

The possibility of exclusion will be especially important in mitigating the adverse effects of increased numbers of states on the prospects for

arise as a consequence of political-institutional factors (e.g., lack of control over exclusion reflected in imperfect property rights). Furthermore, the concepts of jointness and nonexclusion are not analytically distinct since jointness is a necessary but not sufficient condition for nonexclusion (Snidal, 1979). Such distinctions are essential for a more fine-tuned analysis of particular issue areas, although for the more general analysis below we will concentrate on the property of nonexclusion.

¹⁴I use the term "closely related" here because there are some subtle distinctions between the technical and institutional aspects of the relevant restrictions. Within the analysis of market economies, nonexclusion is treated as a technical property of goods (within an otherwise perfect set of property rights), whereas at the international level, problems of nonexclusion may also

international cooperation. If the benefits of cooperation can be restricted to states that behave cooperatively, then such exclusion will make provision in large groups hardly more difficult than in small groups. Indeed, in some sense exclusion makes the problem a small group one of agreement between any single country and the group of already-cooperating states. Admission to the group of benefitting states (or continued membership in the group) will be made contingent upon contribution to the collective effort. Many international organizations involving such diverse issues as trade, money, military alliances, or health regulation are in fact able to practice such exclusion to different degrees. Although difficult bargaining problems over what constitutes a "fair" contribution by various members persist, such exclusionary principles do facilitate substantial levels of cooperation.

Second, consider the impact of asymmetry within a group of states.¹⁵ Hardin (1982, pp. 40-41, 73) demonstrates that the likelihood of cooperation depends not only on the size of the group but on the net benefits received by each cooperating member and hence on the minimum-sized subgroup that can benefit from cooperation. For a given average payoff, a more asymmetric distribution of benefits means that a smaller subgroup is able to benefit from cooperation and therefore that cooperation within the group is more likely. If the issue involves public goods, then other states can get a free ride off this provision (and the distributive implications of the theory of hegemonic stability pertain); if the good is not joint, then there will not be benefits to non-contributors and only the larger states will benefit from their ability to cooperate; and if the good exhibits jointness but exclusion is possible, then noncontributors may be induced to contribute to the cooperative venture (perhaps at a cost they may see as exploitative).¹⁶

Therefore, the general conclusions that cooperation in PD is less likely as group size increases or as asymmetry decreases depend heavily on the exact context of the issue. These conclusions are not universally applicable whenever the issues do not meet the public good conditions of jointness or nonexclusion or both. Moreover, sometimes the cooperation that large numbers serve to impede is only a possibility because there are a large

number of states involved in the issue. Only with these important caveats, can one conclude that *in PD large numbers tend to inhibit cooperation*. In addition, *asymmetries increase the likelihood of some cooperation but the distribution of benefits from cooperation will also be affected by the asymmetry*. Sometimes the distribution will favor smaller states, but in other cases they will favor larger states. Thus generalizations such as the "theory of hegemonic stability" need to be tightly circumscribed by careful specification of the properties of particular issue areas. This is true even within the realm of PD situations, and I demonstrate below that it applies even more forcefully in the coordination problem.

The Role of Time in Prisoners' Dilemma

One limitation of 2×2 games is that their inherently static, one-shot nature is inadequate for analyzing problems where considerations of play through time are important. Although it falls short of a truly dynamic analysis, some understanding of cooperation in a more dynamic world is provided by considering the situation where the same 2×2 game is played (or "iterated") through time.¹⁷ Normally this involves looking only at the impact of future time, but it is also relevant to look at the impact of past and present time as discussed below.

It is now widely understood that the prospect of repeated play into the future increases the likelihood of cooperation in PD even in the absence of centralized enforcement (Axelrod, 1984; Hardin, 1982; Shubik, 1970; Taylor, 1976). Although noncooperation is a rational strategy in single-play PD, repeated play of the game into the future makes it worthwhile to take a chance on cooperating today in the hope that other states will do likewise and mutual cooperation will prevail over the deficient noncooperative equilibrium in the longer term. This possibility is more likely: 1) the longer the time horizon (or expected number of iterations of the game), since, as the final

¹⁵In doing this I implicitly move from ordinal to interval-level payoffs.

¹⁶Smaller states will be willing to join at any cost less than their benefit from participating in the cooperating group. As a result, imposing sufficiently high admission costs on smaller states will have the effect of ensuring that all of the profits of cooperation (i.e., excess of benefits over costs) accrue to the larger states.

¹⁷Iterated games have important limitations as approximations of dynamic situations since, for example, they do not allow for the possibility that the structure of the system itself may change over time. This is an important consideration when we are trying to analyze issues where institutional and political changes may be involved. Ultimately the supergame formed by looking at simple games iterated through time is only slightly more dynamic than the simple game itself (where strategies can be thought of as dynamic insofar as they represent a listing of how the state will act through time). Nevertheless, the supergame does provide an important first step in looking at the impact of playing through time.

game approaches, the incentive to cooperate because of future considerations diminishes; 2) the lower the temporal discount rate or, equivalently, the more future benefits from cooperation are valued relative to the prevailing incentive not to cooperate; and 3) the greater the benefits from cooperation relative to the gains to be had from not cooperating (or the losses to be incurred from cooperating while other states do not).¹⁸ To the extent that these conditions are fulfilled, *the fact that a PD issue is recurring through time will facilitate cooperative behavior.*

If we treat "present" time as the duration of a particular simple PD game embedded within an ongoing issue area, it becomes apparent that the speed with which states can adjust their policies (i.e., the duration of any single PD) has an impact similar to that of a longer future time horizon. The faster that states adapt to each others' actions, the more iterations of the game there are in any fixed length of time. Thus *when states in PD can adapt quickly to cooperative or noncooperative actions by the other side, there are greater prospects for cooperation* for precisely the same reasons that future play improves prospects for cooperation in PD. Improved information and communication—including preparations for policy changes—may be an important aspect of regimes for this reason.

Finally, past experience in PD may either encourage or hinder cooperation because of the mixed effects of learning and building trust between states. This is directly related to the incentives to cooperate because of "reputational" effects that encourage cooperation when games are repeated into the future. Generally, past success at cooperation will positively reinforce cooperation because of the increased reputational costs of reneging on longstanding arrangements. However, past success at cooperation also enhances the stability of a regime so that states may begin to feel they can cheat on cooperation without disrupting the tendency of other states to cooperate. This will be most likely in large number situations involving public goods and is well exemplified by Russett's (1970, p. 99) observation that the tendency of small allies in NATO to try to free ride

on U.S. military provision provides strong evidence of the past success of the alliance.

In summary, the overwhelmingly most important aspect of time in PD is the expected duration of an issue into the future. Present time—as reflected in the speed of adjustment in responding to other states' policies—plays a similar role in facilitating cooperative behavior. Past time plays a relatively small role.

The Coordination Problem and the Coordination Game

The coordination problem is generally less well understood as a problem of collective action and so will be dealt with in some detail here. Sometimes coordination is presented simply as the problem of two or more actors matching policies where they are indifferent about where they match (or have perfectly correlated preferences over alternative matches). A good example is a meeting problem where two friends are indifferent about where they meet for lunch as long as they both go to the same restaurant. Here there is no disjuncture between individual and collective rationality and no problem of collective action. It requires no more than communication and common sense to achieve an outcome that is both individually and collectively optimal.¹⁹

A more deep-seated problem of coordination arises when actors have a strong desire to coordinate but some differences over exactly where to coordinate. This problem is introduced into the meeting problem when friends wish to meet for lunch but each prefers meeting at a different restaurant or at a different time. The normal form of this game is presented in Figure 3.²⁰

This game poses a different problem of collective action than does PD. The problem in PD is that in pursuing its own self-interest, each state imposes costs on the other independent of the other state's policy choice, whereas *in the coordination game each imposes costs or benefits on*

¹⁸The formal conditions for cooperation induced through repeated play are quite stringent and would not appear to offer much help for international cooperation (Taylor, 1976). In large part this is because the purely logical analysis misses some of the important features of the social context which facilitate cooperation, and perhaps even distorts the nature of the problem (cf., Hardin, 1982, chap. 9). Other extensions of the basic supergame model may also enhance possibilities for cooperation (Axelrod, 1984).

¹⁹Two important discussions of the coordination game are Lewis (1969) and Schelling (1960). Their examples of such coordination problems all revolve around problems of information and communication. Although these are certainly important in some international problems (e.g., see Schelling's military examples), they are not widely applicable to issues of longer-term cooperation failures in regimes where communication is possible.

²⁰Figure 3 shows one symmetric version of the coordination problem. Minor variations (and different coordination games) can be created by interchanging the two lowest payoffs in various combinations, although not all of the resulting games are symmetric. However, the basic analytical results are the same as reported here.

Figure 3. Coordination Game

		State B	
		0	1
State A	0	4, 3	2, 2
	1	1, 1	3, 4

the other contingent upon the other's policy. The collective action problem is that neither state can choose its best policy without knowing what the other intends to do, but there is no obvious point at which to coordinate. Easy resolution of the problem (as in the simple meeting problem) is hindered by the inherently opposed interests about where coordination should occur. (State A prefers the northwest cell and State B the southeast cell.) However, once a cooperative solution is achieved, it does not need to be enforced (as did the cooperative outcome in simple PD), but is self-enforcing, because neither state has any incentive to depart from the coordination point in the single-play coordination game. Thus one can appreciate that the problem of coordination is fundamentally different from that of assistance in PD. (Stated more formally, the problem in the coordination game is one of choice between multiple stable and efficient equilibria over which states have opposed interests, whereas in PD the problem is getting away from a single stable but inefficient equilibrium.)

Cooperation in the coordination game depends on the surrounding social-political context. Established traditions or shared principles are likely to provide the basis for the emergence of norms or *conventions* which will guide individual national behavior in an issue area. No centralized enforcement is necessary, because neither state has incentive to depart from an established convention: The prevailing expectations of how the other state will behave which are conveyed by the convention provide a compelling reason for each state to adhere to it in order to gain the benefits of coordination. Any role for centralized authority in coordination problems is likely to be less concerned with enforcement than with codification and elaboration of an existing or latent convention and with providing information and communication to facilitate the smooth operation of the convention. Moreover, because the need for enforcement that exists in PD is absent, in coordination states will be more resistant to surrendering autonomy to a central agency because an alter-

native decentralized resolution is available. Finally, unlike PD where an obvious "cooperation point" exists, a sovereign would have no clear criterion (independent of conventions) upon which to base a choice between alternative coordination points. Therefore the simple coordination problem is almost certain to result in a decentralized solution.

Although this simple model of coordination applies to an important class of international cooperation problems (e.g., standardization of measurement systems, standardization of equipment within NATO, or recent European attempts to adopt common standards in telecommunications equipment), it is far too restrictive and unrealistic for many other important problems of international coordination. Consider the nature of attempts at international economic cooperation reflected in the recent string of economic summit meetings between leaders of the largest capitalist economies. In essence, the shared objective has been to agree on broad macroeconomic objectives and to coordinate policies in order to achieve them.²¹ Two important differences distinguish this situation from the simple coordination game presented in Figure 3: 1) available policies are neither dichotomous nor discrete but involve states choosing over a broad policy continuum (e.g., from "contractionary" to "expansionary"); and 2) coordination is not an all-or-nothing problem and there is no need to meet exactly, provided that policies fall within a range of compatibility. Instead, states might be better characterized as preferring to coordinate policies as closely as possible where, in general, closer is better, *ceteris paribus*. However, at the same time different states will prefer to coordinate at different points (e.g., the West Germans may have a relatively greater concern with price stability than the French) and so some opposition of interests remains within the overall desire to coordinate.

²¹Of course, many other international economic issues—for example, beggar-thy-neighbor trade policies—will fall into the PD category of collective action problem. But problems such as demand management are closer to the coordination problem. For a discussion of this see Whitman (1977) and Gordon and Pelkmans (1979, part II). Pelkmans's argument is particularly relevant in distinguishing the different nature of international economic problems of what he calls "foreign economic policy" such as trade policy and (international) "demand management policies." He argues that the former require mutual assistance (or "negative cooperation") in the form of mutual restraint from policies that hurt others; the latter require coordination (or "positive cooperation") and meshing of policies between states. However, he fails to draw the corresponding analytical distinctions and often seems to be treating both problems as equivalent to PD.

This conception of the coordination problem provides interesting conclusions for the nature of regimes as I turn to extensions of the simple coordination game.

Multiple Choices in the Coordination Game

A modified version of the standard 2×2 coordination game is presented in Figure 4. The locus ACB is the axis of perfect coordination of policies between two states. However, there are differences over where to coordinate so that State A prefers points on the axis which are closer to A where the payoff is (4,3), whereas B prefers coordination outcomes closer to B where the payoff is (3,4). An intermediate point such as C where the payoff is (3.5, 3.5) might provide a compromise coordination point which would ease resolution of the coordination problem in the absence of a convention (but might complicate it if a different convention already existed). Nevertheless, a point like Q, which is not a point of perfect policy coordination, might be preferred to C by State B because it is sufficiently closer to B's most preferred outcome at B to compensate (from B's perspective) for the deviation from the coordination point. Thus bargaining over the cooperative outcome need not be restricted simply to points on the coordination axis, is as demonstrated below. (However, I assume that for every point Q off the coordination axis there is at least one point on the coordination axis, Q^* , which is preferred by both states; that is, the coordination axis is the core of the coordination game.)

Because the graduated form of coordination games is more complicated than that of PD games, it is instructive to derive it using national indifference curves based on assumptions about the nature of coordination problems.²² Because a

²²Technically, the greater complication arises directly from the nonseparability of coordination games. The

Figure 4. Coordination Axis in the Coordination Game

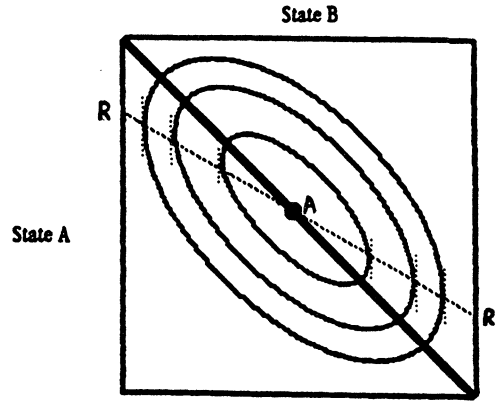
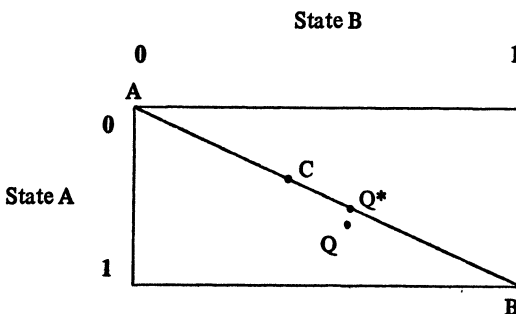


Figure 5. Indifference Curves and Reaction Curve for State A

Point A is State A's most preferred outcome and the center of the indifference ellipses that stretch along the coordination axis (i.e., ACB in Figure 4) which is their principal axis. State A's reaction curve (broken line $\overline{RR}$) shows A's optimal choice *given* any choice by B. It is found by locating the point of tangency of any policy of B (represented by the dotted vertical lines) with the indifference ellipse closest to A. This provides the locus of A's optimal response to any policy choice by B.

state's utility depends on both the distance of the outcome from its most preferred outcome and the distance of the outcome from the coordination axis, its indifference map is reasonably represented by a set of elliptically shaped indifference curves. These curves are concentric to the state's most preferred point and stretch along the coordination axis as shown in Figure 5.²³ Ellipses closer to the center represent higher utility levels, while the more concerned a state is with coordination (relative to its concern for proximity to its optimal point), the more elongated and less circular will be its indifference ellipses.

An analysis of these indifference curves shows that despite the importance of coordination in determining a state's welfare, some incentive to depart from any coordinated outcome may persist. In Figure 5, the broken line $\overline{RR}$ is a reaction curve showing A's best policy given any choice by

graduated version developed below is not unique but is the substantively most appropriate and fruitful (Snidal, 1981, chap. 3).

²³If utility is a linear function of the two distances squared, then the indifference curves will be proper ellipses. Otherwise, roughly elliptical curves provide a useful representation for this sort of coordination problem.

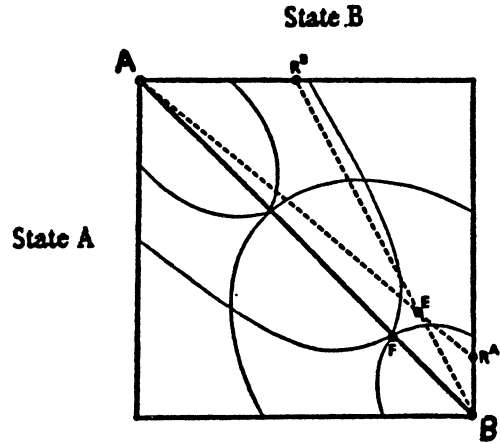
State B. Except when B has chosen that policy which corresponds with A's optimal outcome (i.e., at the center of the ellipses) State A has an incentive to depart slightly from the coordination locus in order to get closer to that optimal point (although balancing this against the cost of increased departure from the coordination axis), and to increase its overall utility. State B will have a similar tendency to depart from the coordination axis, and its reaction curve can be derived in a like manner.

Figure 6 presents the joint indifference map for the two states and their corresponding reaction curves. Each state has a most preferred outcome at one end of the coordination axis and elliptically shaped indifference curves along the axis. (There is no need to show the coordination axis beyond these two points because neither state would prefer coordination points outside this range to ones available between their respectively optimal points.) The sets of indifference curves are tangent to one another along the coordination axis which represents the set of core (or Pareto efficient) outcomes. The two broken lines are the reaction curves which show how states would adjust to each other. Their intersection at E represents a (Cournot and Nash) equilibrium outcome. It is striking that this equilibrium point is not on the coordination axis. Instead there is a stable equilibrium which is deficient since there are other outcomes (e.g., at F) that would be preferred to it by both states.²⁴ Inspection of the indifference maps shows that E is in a PD relationship to F.

This provides a significant insight to the nature of coordination games: *Coordination is a fundamentally different problem when viewed in the small than when viewed in the large.* Indeed, coordination is an inherently large-scale problem which "disappears" when examined in terms of the smaller-scale policy adjustments observed in the graduated game. But the fact that the problem in the small within the coordination game is actually PD does not mean that it can be dismissed as a separate problem or subsumed under PD. Instead, coordination has fundamentally different implications for the formation of regimes which now have to manage simultaneously the

²⁴The Cournot behavioral assumption (that each state assumes the other will not respond to changes in its policy choice) has severe limitations in a game situation when we are considering strategic actors. However, the basic nature of the graduated coordination is independent of this assumption, which serves only to illustrate the sorts of deficient outcomes that can occur. Other assumptions incorporating more sophisticated notions of dynamic bargaining behavior (e.g., that both sides behave as Stackelberg leaders) generate even more perverse and unstable outcomes.

Figure 6. Joint Indifference Map and Reaction Curves for States A and B



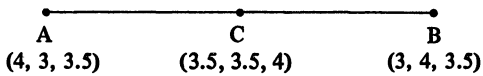
Lines $\overline{AR}^A$ and $\overline{BR}^B$ are reaction curves for States A and B respectively. Their intersection at E is a stable (Cournot) equilibrium. The deficiency of E can be seen by observing that a point such as F lies on a higher indifference curve for both states. To see that E is in a PD relation to F, construct the subgame defined by the rectangle of which E and F are opposite corners.

need for coordination in the large and the incentives to depart from perfect coordination in the small. Thus the meeting problem of the 2×2 coordination game which easily could be resolved by communication and reliance on conventions becomes much more deepseated in the graduated game. Further differences and complications emerge as one investigates other extensions of coordination games.

Multiple States and Asymmetry in Coordination Games

The extension of the coordination game to the n -actor case is not as straightforward as the corresponding extension for PD. The simple 2×2 game can be expanded to a symmetric n -dimensional cube where there are n coordination points, each of which represents the outcome preferred by one state. (A fanciful example might be the choice of a single global language.) Although such coordination problems are conceivable, they do not correspond to many substantively important situations as discussed above. A simpler and more useful representation of the multi-actor coordination problem can be developed from the coordination axis in Figure 4. Here states see coordination not as a matter of agreeing between separate and unrelated coordination points but as a problem of meeting on a coordination axis (located in the

Figure 7. Asymmetry of Three-State Coordination



n -space spanned by their individual policy-choice dimensions) with each having its most preferred outcome somewhere along that axis. Again, each state's utility depends on the closeness of the outcome to its optimal point and on the closeness of the outcome to the coordination axis.²⁵

A first difference from PD arises with respect to asymmetries among states. Whereas the n -state extension of PD could in principle be fully symmetric, the extension of coordination is necessarily asymmetric. Consider the coordination axis ACB of Figure 4 when I add a third state whose preferred outcome lies between A and B, say at C, and whose payoff is shown as the third element in the payoff set. The coordination axis now looks like Figure 7. The new state will be advantaged relative to the other states (*ceteris paribus*) since it will necessarily receive an outcome with an ordinal ranking of at least 3.5 (and probably higher), whereas at most one of the other countries will do as well. States that happen to have most preferred outcomes closer to the middle of the coordination axis are in an inherently better position than states with more extreme preferences. Thus coordination problems are almost certain to put states in nonsymmetric relations with one another.

A second source of asymmetry arises when some states have more impact on the outcome than do others. In PD this typically arises because some states impose relatively greater costs (or benefits) on others through their choice of policies. In coordination, the issue is not one of imposing externalities but rather one of the relative need to coordinate. The policy choices of some states will be inherently more significant than those of others and thus provide a relatively greater need for coordination. In general, the need for coordination between any two states will vary directly with their levels of interdependence upon one another. For example, the typical advanced industrial country will find coordination with the largest industrial powers (United

States, West Germany, Japan) more important than coordination with the smaller economies (Denmark, Canada, Netherlands).²⁶ This is likely to give larger countries an advantage in negotiations over international coordination.

The advantage of larger states in international coordination problems is further reinforced to the extent that they are less interdependent with other states and hence less reliant on coordination. This point is illustrated in Figure 6, where State B is substantially less concerned with coordination than is State A, as is reflected in its more circular and less elongated indifference ellipses. As a result, the noncooperative (Cournot) outcome of this situation at E is substantially closer to State B's most preferred outcome than to State A's. It is reasonable to suppose that this would constrain bargaining over cooperation so that any fully cooperative outcome would fall at a point on the coordination axis which also favored State B (e.g., point F). Thus, unlike PD where asymmetries sometimes favor smaller and less influential states, *in coordination asymmetries favor larger and more influential states*.

The impact of increasing the number of states involved in a coordination problem is also somewhat different from the case of PD. Again, some of the social-political features of large versus small groups which make organization difficult in PD have a similar effect in coordination. To the extent that large groups impede communication and complicate bargaining, this will make agreement on a coordinating convention more difficult unless tradition or particular aspects of the situation offer an "obvious" (focal point) solution. However, the consequences of reduced visibility of the individual actions of states in larger groups will not be as serious in coordination as in PD. Although it will give individual states an opportunity to depart slightly from the convention (i.e., in the small as represented by the PD zone in Figure 6), the nature of the coordination problem is such that this will not be a problem in the large since national incentives are to coordinate volun-

²⁵This is a drastic simplification. The coordination axis might not be a single line but a plane or hyper-plane in n -space. The assumption implies that all states perceive the issue as involving a single dimension of coordination and that their indifference curves are ellipsoids with the coordination axis as principal axis.

²⁶As always, such generalizations need to be treated with caution and appropriate attention to particular contexts. For example, the United States may have a relatively greater need to coordinate with Canada than with larger European states because the two North American economies are so closely intertwined. This is also a good example where the need to coordinate is not symmetric, because the Canadian need to coordinate with the United States greatly exceeds any reciprocal American need to coordinate with Canada.

In terms of the ellipsoid-shaped indifference curves, greater need for coordination with a country is reflected in a shorter minor axis in the plane determined by the two countries' policy dimensions.

tarily with other states. Thus larger numbers of states will give rise to communication and other problems of organization in coordination but not to the strategic incentives to act uncooperatively which are pervasive in frustrating cooperation in *n*-state PD.

Indeed, an increased number of states will reduce the likelihood of the breakdown of cooperation resulting from the deliberate actions of states. Once a coordination point is established, there may be a strategic incentive for each state to depart (or threaten to depart) from the coordination point intentionally in order to force others to accept its preferred coordination point or none at all. However, as the number of states involved in a coordination problem increases, then the impact of any single state and hence the likelihood that it could blackmail the others in this way decrease dramatically. Thus, *in coordination, larger numbers of states do not impede cooperation to nearly the extent they do in PD and facilitate cooperation in some cases.*

Two final points are relevant to the question of increasing numbers of states and asymmetry in coordination. First, exclusion does not play an important role (as it did in PD) for the simple reason that it is not a relevant option in coordination. In PD, costs and benefits are imposed independently of the other's actions so that exclusion (if possible) can affect one state without affecting another. In coordination, costs and benefits are imposed contingent upon the other's actions so that the other cannot be excluded without in some sense excluding all. Thus the very essence of the coordination problem is that exclusion (or free riding) is neither feasible nor desirable. Second, whereas in PD asymmetry among states facilitates cooperation by lessening communication and bargaining problems among the most relevant set of actors, in coordination it decreases cooperation by increasing the strategic incentive to bargain over the coordination point. The exception is when a single state is sufficiently dominant that it can unilaterally determine the coordinated outcome. Moreover, unlike some cases of PD, large countries will tend to benefit more from these agreements since they will have a greater voice in determining the point of coordination. Of course, small countries will be better off than if no coordination was achieved.

The Role of Time in Coordination Games

Because of the inherently stable nature of single-play coordination, playing through time has much less impact than it does in PD. But to the extent that they are important, the three different aspects of time—past, present and future—have substantially different impacts on coordina-

tion games than on PD. The greatest similarity of impact is with respect to the length of time that the game has been played in the past. Because cooperation in coordination depends heavily on the ability to agree on common conventions or norms, *past experience* with and learning of norms and conventions *will be very stabilizing for regimes*. The incentive for states to cheat slightly on the coordinating convention in the small may be enhanced by this greater stability of the regime. However, states would have to weigh these smaller gains against the impact cheating might have in diminishing the clarity of well-established norms and hence of upsetting the much greater benefits of coordination in the large.

The role of future time will be the exact opposite of that in PD. In PD the prospect of repeated play provides states with an incentive to cooperate that they do not have in a one-shot situation. In coordination, the single-shot outcome is likely already to be a cooperative one because the ability to coordinate somewhere is more important than are relatively minor differences about where to coordinate. A longer duration of the game gives states incentives to be more concerned with the exact distributional consequences of particular coordination outcomes. In the extreme, it may induce them deliberately to upset prevailing coordinated outcomes in an attempt to institute a movement to other conventions which are more favorable to them. The prospects for such strategic action clearly rest on the (discounted) relative value of future versus present outcomes and the perceived prospects of implementing a change. Thus *playing through time may be destabilizing in coordination problems*, but these considerations will still typically be dominated by the overall stability of the coordination situation.

The impact of present time follows directly from a consideration of the incentives to upset a prevailing coordination outcome for strategic reasons. If adjustment of policies is a lengthy process, then attempts to initiate change in coordinating conventions will necessarily entail greater costs from a longer period of noncoordination during the adjustment process. In general, this should discourage states from trying to change the convention and thereby enhance the stability of the regime.

Two Different Regimes: Coordination and Assistance

Table 1 summarizes the main differences between the problems of collective action posed by coordination and PD. The entries are necessarily simplified and do not capture all of the nuances that may occur within each basic game situation according to particular contextual differences. As

Table 1. Summary of Key Differences and Similarities between PD and Coordination

	Prisoners' Dilemma	Coordination
Basic problem	Assistance: states interdependent because each imposes costs on the other independent of the other's actions	Coordination: states interdependent because impact of each state's choice contingent on other's choice
Basic solution	Contract	Convention
Effect of graduation	Strategic structure unaltered: game is same in small as in large	Strategic structure altered: game is fundamentally different in small than in large
Effect of multiple players	Generally inhibits cooperation: alleviated by asymmetries and possibility of exclusion	May inhibit formation of regimes; enhances stability of regimes
Effect of asymmetries	i. Facilitates cooperation ii. Without exclusion favors smaller states; with exclusion favors larger states	i. Slightly facilitates cooperation ii. Favors larger states
Impact of time	Future time and rapid adjustment increase prospects for cooperation	Past time increases whereas future time and rapid adjustment decrease prospects for cooperation

a result, the differences between the two situations are drawn somewhat more starkly than they may actually appear in real world situations. Nevertheless, the table usefully summarizes the fundamentally different natures of the two problems of collective action.

In comparing these two problems of collective action, we have also (directly and indirectly) been comparing the problems each poses for international cooperation and the implications for resolution of these problems through international regimes. For example, the discussion of the impact of graduated policy options or of increasing numbers of states each had important but different consequences for the prospects of cooperation in the two situations. Further differences can be expected to occur within each of these two broad categories of collective action problems according to various contextual circumstances such as asymmetries among states or prospects for future play. These differences will be amplified by historical and other differences not directly accounted for in the game theoretic analysis. Moreover, these are only two of a much larger set of possible situations of interdependence and other possible strategic interrelations among states may have yet different implications for regimes.²⁷ All this suggests an enormous potential diversity of regimes. However, I concentrate on the diversity that arises from the two polar cases

of coordination and assistance problems and the corresponding implications they hold for regime characteristics—especially as they pertain to questions of institutionalization and the stability (and adaptability) of different regime institutions. This approach demonstrates quite clearly how different problems of collective action may result in different regime solutions.

The most basic distinction between the two categories of regimes is that between contract and convention as forms of solutions to cooperation problems. This distinction is often treated as one between the need for centralized enforcement versus the possibility of decentralized outcomes. Although this is the case in many circumstances, it is not a hard and fast rule. As I have discussed, cooperative outcomes to PD sometimes can be achieved without centralized enforcement when a problem recurs through time; similarly, centralized institutions could provide the basis for conventional outcomes. Instead of being viewed as a defining characteristic of the two situations, the occurrence of centralized versus decentralized resolutions is better treated as a consequence of the underlying nature of solutions by contract and by convention. Contract is a solution to a particular type of cooperation problem of assistance (i.e., separable externalities) and is based on the mutual performance (or forbearance) of particular acts so as to benefit others. States surrender some of their authority or autonomy in return for other states doing the same. For example, states might agree to surrender the right to produce certain sorts of weapons or to impose certain restrictions on imports. Convention is a solution to a different problem of cooperation (i.e., non-

²⁷As mentioned in note 7, a prominent case in point is "Chicken" which, for example, contains interesting problems of escalation not found in either PD or coordination.

separable externalities) and is based on the coordination or harmonization of policies among states. It does not require that states give up any autonomy per se although negotiation of coordination points will affect the choices they will want to make independently. For example, states might agree on target macroeconomic policies and for that reason find it in their self-interest to pursue those targets even though they are still free to pursue other targets. Thus contract and convention provide different categories of basic solutions to fundamentally different problems of interdependence. As Stein (1982, p. 316) points out, cooperation in such issues depends on "rational self-interested calculation (which) leads actors to abandon independent decision-making in favor of joint decision-making." But the underlying differences in the two situations result in important differences in the nature of joint decision making and hence in the nature of regimes.

Institutionalization

A first distinction concerns the nature of the basic rules and the extent of institutionalization (i.e., the nature of formal political arrangements) which characterize an issue area. In convention, rules will be largely self-enforcing and states will obey them (which is in the collective interest) because it is in their own self-interest. In contract, rules are not self-enforcing and there will be a tendency for them to be weakened by states cheating. The conflict between self-interest and collective interest persists even after rules are formulated. This difference between convention and contract will be reflected directly in the level and nature of institutionalization and, as mentioned above, PD regimes that rely on contract will likely be more centralized in order to regulate the actions of states within agreed-upon rules.

Regimes pertaining to coordination problems, and based on convention, will generally have low levels of institutionalization concerned primarily with facilitating the choice, interpretation, and observance of a particular convention. This will involve information gathering and informal consultation about the preferences and policies of states as well as providing a forum for the resolution of bargaining problems pertaining to the choice of a particular convention. Enforcement in the large will not be necessary, and exclusion does not make sense given the nature of the problem. (However, a subset of states may try to exclude others from the process of selecting a convention in order to impose one that corresponds more closely to their own preferences.) Some local enforcement may be desirable to prevent minor deviations from the convention in the small. States may be willing to accept this enforcement

although adherence may be achieved through (negotiated) voluntary restraint since it entails only a small sacrifice to reinforce a convention producing much larger benefits.

Contract solutions to PD will likely require much higher levels of institutionalization in order to overcome the continuing incentives for states to defect from cooperative arrangements. This will be reflected in formal agreements restricting national behavior (i.e., akin to Hart's primary rules of law) as well as procedures to support these rules through centralized processes of adjudication (i.e., akin to Hart's secondary rules). Needless to say, the centralized powers of sanction are typically very limited and contract is sustained more by needs of maintaining ongoing cooperation and reputational considerations than through fear of any international sovereign. One notable exception to this is the sanction of exclusion from a regime. Threat of exclusion, if credible, may be an important device for ensuring that states behave cooperatively. The credibility of the threat will depend in part on the properties of the particular issue area (e.g., impossible for many international commons problems but not for trade issues) and in part on the nature of the political arrangements (e.g., a military alliance able to determine precisely the scope of its membership). If such exclusion is possible, then states may accept the authority of the international regime rather than face exclusion. More typical than the use of threats of expulsion against existing members is the requirement of acceptance of regime rules as a condition prior to admittance into the regime. This probably constitutes the single most important source of "centralized" enforcement.²⁸

The second source of cooperation in assistance problems is the noncentralized enforcement that arises from reputational considerations and the prospects of playing through time. Indeed, one consequence of playing through time (or, alternatively, of the reaction time to each others' policies becoming exceedingly rapid) is that the strategic structure of the situation is transformed into one similar to that of the coordination game (Snidal, 1981; Taylor, 1976).²⁹ To the extent that

²⁸Sometimes countries are able to appeal to an outside third party to enforce an agreement. One example has been reliance on the superpowers to supervise regional agreements. This has been one important role for the United States in the Middle East over the past decade.

²⁹This observation opens up important considerations for understanding how different solutions are possible for the same problem of collective action. Except for brief comments in the present and next paragraphs, I will not investigate that here. For a discussion of this, see Hardin's (1982) discussion of "contract by convention" as well as Snidal (1981).

these other incentives are adequate for maintaining cooperation, then solution by contract and centralized enforcement are not necessary. However, uncertainty about the possibility of maintaining cooperation over the long run, and the lack of trust among states coupled with continuing incentives to defect in any single play of the game (especially in issues involving large numbers of states), may make some form of centralized monitoring efficacious for maintaining cooperation. Therefore, even in this situation of repeated play through time, a greater level of enforcement is required (in PD) than in the pure coordination situation.

However, reputations and considerations of ongoing cooperation are not restricted to single-issue areas but are important across issue areas (cf. Hardin, 1982, pp. 186-187). A horizontal proliferation of PD games may provide a more important incentive for cooperation than the vertical continuation of any single PD issue area through time. Fear that noncooperation will spread into other issue areas provide incentives for states not to succumb to immediate temptations to defect for short-run, issue-specific benefits. Indeed, this is a much more important aspect of issue-linkage than the more often discussed exchange form of linkage based on the linking of issues in a bargaining context. It is directly related to broader questions about the emergence of an international society with multiple overlapping interests and concerns. To the extent that such a society is emerging, the individual PD situation becomes embedded in a broader social context, and cooperation is increasingly possible with less formal centralized enforcement. In the extreme, variation between issues according to their strategic structure would tend to be dominated by the similarity imposed by these broader social considerations. But in less fully articulated international societies such as the present one, important regime variations will persist. Social considerations may be sufficient to enforce cooperation in PD issue areas of relatively minor importance where costs and risks of cooperation are low relative to benefits (e.g., international copyright law) but not in vital issue areas where they are high (e.g., military cooperation). Thus, in the current situation, successful resolution of major PD problems will require a higher level of institutionalization than in coordination problems.

Regime Stability and Transformation

Because a comprehensive international society is not fully developed, questions pertaining to the stability of regimes within particular issue areas are important. Sources of instability could include any of a wide range of changes in the circum-

stances or context of an issue area, but two potential sources of instability stand out as being of particular interest. The first is the stability of the regime in the face of underlying changes in the issue area itself.³⁰ The second is the stability of the regimes and their rules also to change. Second, issue power structure or, alternatively, increasing or decreasing asymmetry among states. But in analyzing each of these potential sources of instability one must be somewhat wary of the presumption that stability—even stability of cooperative outcomes—is always a virtue. First, changes in an issue area may affect the nature of desired cooperative outcomes so that it is desirable for regimes and their rules to also change. Second, different states may have different evaluations of the outcomes produced under various regimes, and those relatively least advantaged under particular regimes may welcome change. Of course in other cases the stability of cooperative outcomes will be more widely viewed as beneficial.

The striking feature of the coordination game is its stability in the single-play version, which is only slightly decreased when the situation recurs through time. If issue change is so substantial that the prevailing convention becomes dysfunctional (i.e., worse than no coordination) for regime members, then the regime will dissolve or transform. But for less dramatic issue change (which leaves the prevailing convention suboptimal but still better than no coordinating agreement whatsoever), the stability of the regime combined with likely disputes over which new convention to adopt will make it difficult to initiate a change to a better convention. This stability means that the regime will not always be sufficiently flexible in responding to new conditions which affect an issue area. This problem will be difficult to resolve because by their nature conventions cannot change quickly if they are to be effective. It can be mitigated through greater reliance on centralized institutions to adapt conventions to particular prevailing issue circumstances, but this negates many of the other advantages of conventional solutions which resulted from decentralized regimes. Alternatively, the existence of a single or small number of dominant actors may enable them to act as leaders in adapting new coordination points to changes in underlying conditions. However, except for this role of leadership, a

³⁰This sort of change must be limited to changes that occur within the basic strategic structure of the game. For example, an increase in the external military threat might increase the optimal level of security provision in an alliance or a change in economic conditions might require some common shift in economic policies. But changes so dramatic as to alter the underlying strategic structure cannot be analyzed in the same way.

decrease in the dominance of a single state is unlikely to be destabilizing to coordination outcomes. Therefore, coordination regimes are likely to be stable but, in the absence of leadership, may suffer from a lack of adaptability as issue circumstances change.

The situation in PD is very different. The inherent instability of PD outcomes will be aggravated by changes in issue areas which affect the distribution of costs and benefits of cooperation—and perhaps even the precise definition of cooperative behavior by the individual state. Because it can be profitably abandoned by the individual state, success and stability of a regime depend on its ability to deliver benefits from cooperation relative to those available from alternative regime arrangements. This adjustment will be facilitated by the higher degree of institutionalization associated with contract regimes which enhances their capacity for adaptation and transformation. Centralized institutions will provide a framework for modifying the contract while a past history of cooperation will facilitate the emergence of new arrangements. But, especially with regard to larger adjustments, these transformations will be controversial since states have substantial conflict of interest in PD problems. Bargaining over the exact nature of new regimes will be intense and only well-established contract regimes will be able to orchestrate a successful transition without disruption of cooperation. The exception to this rule will be the special case of regimes concerned with public goods provision where leadership is provided by a single dominant actor.

Changes in the Distribution of Power among States

The impact of changes in the distribution of power among states can be usefully discussed in terms of the theory of hegemonic stability. The first (static) claim of the theory that the prospects for emergence of cooperative regimes are likely to be enhanced by asymmetries among states is essentially supported in both PD and coordination situations although it is subject to some refinement. In PD, the applicability of the generalization will depend heavily on the possibility for exclusion. If there is no exclusion, then we are in the public good case (which is the analytical underpinning of the theory) and the importance of dominant members to regime stability is explained by Olson's "privileged group" analysis. Nevertheless, a problem of cooperation will remain because the dominant member will not provide the collectively optimal level of the good except in concert with other states. If exclusion is possible, then cooperation may be possible without a dominant state although such asymmetry

will likely facilitate cooperation. In the coordination problem, conventions are likely to emerge without dominant states which are therefore not as important to cooperation as in PD. However, cooperation may be facilitated through the prominence of dominant states is establishing conventions—especially if changing circumstances require modification in the convention. Therefore, the claim of hegemonic stability theory about the role of dominant states in facilitating cooperation is largely valid, although it should not be over-stated to suggest that hegemony is necessary for, or always leads to cooperative outcomes.³¹

The second (static) claim that regimes based on the leadership of a dominant state will favor small states is not generally valid. The case where it does apply is again that of international public good provision where exclusion is impossible. But if exclusion is possible in a PD situation, then small countries will not have a free ride, and there is no reason to suppose that they will bear less than their fair share of the costs of cooperation through the regime. Finally, the claim is simply wrong in the coordination problem where regime outcomes will favor larger states which have a greater influence in determining coordination outcomes as discussed above.

Strictly speaking, a static model is inadequate to assess the dynamic claim that international regimes will emerge and decline in response to shifts in the concentration of power in an issue area. (The same applies to the derivation of hegemonic stability arguments themselves.) However, comparative statics analysis allows some observations about the reasonableness of the claim in terms of the preceding discussion of regime stability and transformation. Again, the theory seems plausible for the case its derivation is based upon, public goods, where the decline of a dominant power should decrease the regime's success in providing public goods until the emergence of a new dominant state restores the group's privileged status. In PD with exclusion or centralized enforcement, system stability will not be so dependent on the existence of a dominant state, and the regime may well be able to outlast shifts in the underlying issue power structure. In coordination problems, the decline of a dominant state need not herald the decline of the regime. The stability of the situation is such that the regime may persist indefinitely. Indeed, the most important role of dominant states is not that of preserving the status quo but of providing leadership in transforming the regime. Therefore,

³¹See Keohane (1984) and Snidal (1985) for further elaboration and qualification of this claim.

in coordination, the greatest challenge posed by the decline of a dominant state is not instability but the decreased ability of the regime to adapt to changes in the issue area.³²

These dynamic considerations are quite speculative and a more precise specification of how states behave through time would be needed to obtain clearer predictions. However, even these crude speculations raise doubts concerning the general applicability of hegemonic stability theory and indicate the dangers of failing to distinguish between different problems of collective action and cooperation. Thus, the important point is not that the theory is right or wrong, but that it is inadequately specified to be taken as a general theoretical claim. Leadership in the international system is important, but it is important in different ways in different circumstances. The blessings of leadership are more mixed when viewed from the different perspectives raised by different problems of collective action.

Conclusion

These simple game models of PD and coordination are very spare and necessarily have severe limitations for the analysis of complicated real world issues. But even though they cannot capture all of the details and contextual nuances of particular issues, they provide valuable insights into the complexities posed by strategic interactions among states and how these affect international cooperation.

The analysis above has shown the great diversity of problems of international cooperation and the diversity of regime characteristics which is likely to result. The contrast between PD and coordination demonstrates that simple generalizations based on the PD model cannot be uncritically generalized to other situations. PD (and public goods) is an important problem, but it is not the singular problem of international cooperation, and other models such as coordination better characterize some important issues. These different strategic structures, and different con-

textual variation within strategic structures, have important consequences for regime characteristics ranging from the degree of centralized authority they require through their relationship to the distribution of global power and interest. Thus the analysis of the formation and development of international political regimes cannot be studied without an appropriate understanding of the strategic structure of the underlying issue area.

These same conclusions apply to the study of political organization and collective action more generally. Olson's (1965) public good analysis has been taken too literally as *the* problem of collective action. In fact there are many problems of collective action, and public goods are only one interesting category. Many other problems are PD situations involving purely private goods, while others, such as coordination, are even more fundamentally different. Recognizing these differences and coming to grips with them in various areas of political analysis will improve our understanding of the role of political organization in resolving collective action problems and help to explain the variety of institutional forms that such organization can take.

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³²Further comparisons between PD and coordination with regard to their stability and adaptability can be made in terms of the distinction between "changes within regimes" and "changes of regimes" (Krasner, 1982a, pp. 187-188). Centralized institutions allow for a cleaner separation between fundamental principles and operating rules than do decentralized regimes, where the two may not be easily distinguished. Therefore centralized institutions will be better adapted to facilitating changes in regimes to deal with changes in issue circumstances without upsetting basic regime principles, and regime adaptability and stability may cease to be conflicting characteristics and become complementary.

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